

BIE 20103: REQUIREMENT ENGINEERING

Software Requirements Specification

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) defines the functional and non-functional requirements for the Azura House Automation System. This document is intended for stakeholders including restaurant owners, managers, staff, developers, and academic evaluators. This document serves as a baseline agreement of system behaviour before design and implementation.

1.2 Scope

The Azura House Automation System is a web-based system designed to automate ordering, payment, and inventory management for a small local restaurant.

The software product is capable to:

i) Allowing customers to place orders through QR-code based digital menus.

ii) Processing payments digitally and recording transactions automatically.

iii) Managing inventory with real-time stock updates and low-stock alerts.

The software product is not capable to:

i) Handling payroll and employee salary management.

ii) Performing advanced accounting or tax reporting.

The user of this system will benefit:

i) Reducing manual work and human errors.

ii) Improving order accuracy and customer experience.

The objectives of the system are:

i) Automate restaurant workflows end-to-end.

ii) Improve operational efficiency and reporting accuracy.

The goals of this system are shown in Figure 1.

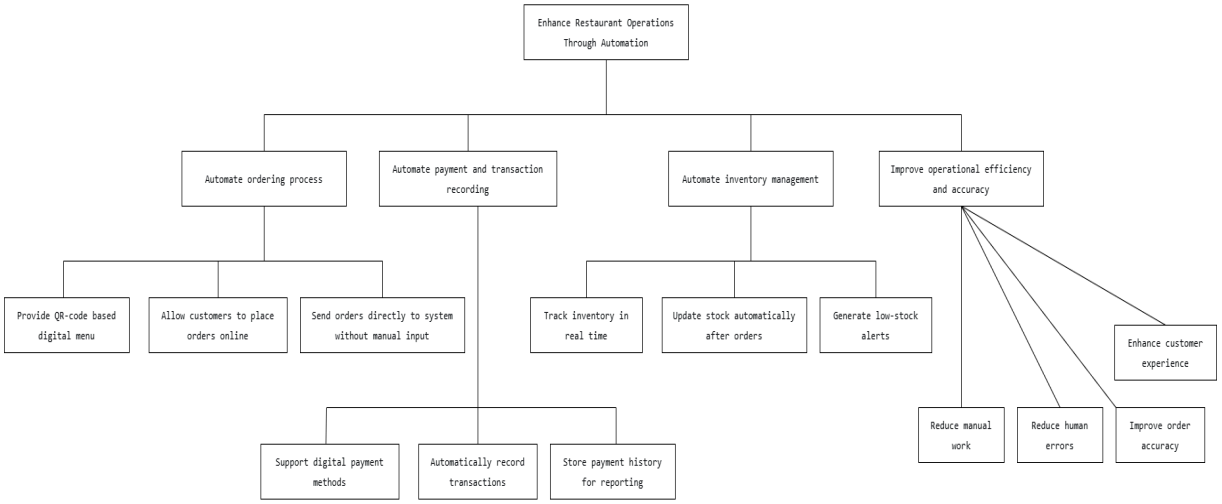


Figure 1: Goal Model

1.3 Definitions, Acronyms and Abbreviation

- SRS: Software Requirements Specification
- URD: User Requirements Definition
- KDS: Kitchen Display System
- QR Code: Quick Response Code
- Actor: A user or external system interacting with the system
- UC: Use Case

1.4 References

- Assignment 1: User Requirements Definition (URD) for Azura House Restaurant (assignment 1 re.pdf)
- Project Title Form – Automating Workflow for Small Local Restaurant (titleformgroup5.pdf)
- BIE20103 Requirement Engineering Project Instruction Document (bie20103projectinstruction0311.doc)
- IEEE Std. 830-1998 – Recommended Practice for Software Requirements Specifications
- As-Is and To-Be Process Models – Azura House Restaurant (Appendix A of Assignment 1)

1.5 Overview

This Software Requirements Specification (SRS) document is organized into several sections to clearly define the requirements of the Azura House Cafe Automation System.

Section 1 provides an introduction to the system, including the purpose, scope, definitions, references, and an overview of the document structure. It outlines the objectives, capabilities, and limitations of the proposed system.

Section 2 describes the specific requirements of the system. This section includes detailed use case specifications, activity diagrams, and sequence diagrams that illustrate system behavior. It also presents the domain class diagram, business rules, and requirement traceability matrix to ensure all requirements are properly mapped and validated.

Together, these sections provide a complete and structured description of the system requirements to support design, development, and evaluation activities.

2. SPECIFIC REQUIREMENTS

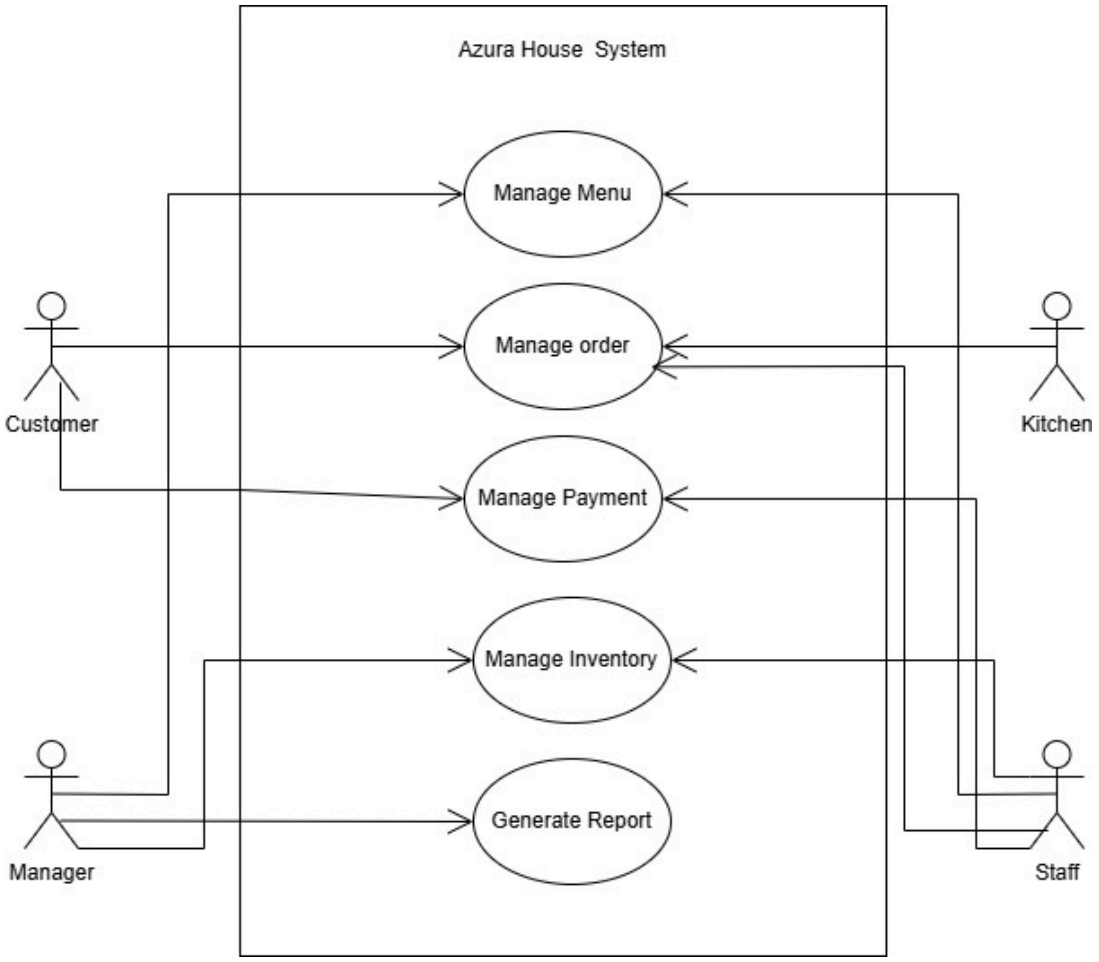


Figure 2 : Use case diagram

2.1 Manage Menu use case specification

History Log	1.0.0	1. Initial creation of Manage Menu use case	
	1.1.0	1. Added alternative flows for add and delete menu	
	2.0.0	1. Updated use case to align with activity and sequence diagrams	
Version	2.0.0		
Use Case ID	UC-1		
Use Case Name	Manage menu		
Created By	Muhammad Faiz Ikhwan Bin Kamarudin	**Updated By**	Muhammad Amirul Hafiz bin Jasmani
Date Created	1/15/2026	**Last Revision Date**	1/17/2026

Actors	Manager and Staff
Description	This use case deals with the way in which the manager or the staff performs the operation of adding or deleting menu items in the Azura House Automation System. This is done by validating the menu details to update the menu records.
Preconditions	1. Manager or staff has successfully logged into the system 2. The Azura House Automation System is online and operational 3. Manager or staff has authorization to manage menu items
Post conditions	1. Menu records are successfully updated in the system 2. Changes are reflected in the digital menu 3. A success message is displayed to the user
Normal Flow	1.0 Manage Menu 1. Manager or staff selects the Manage Menu option 2. System displays menu management options 3. Manager or staff chooses to add a new menu item 4. Manager or staff enters menu details 5. System validates the menu input 6. System saves the menu details into the database 7. System displays a successful message 8. Use case ends
Alternative flow	A.1 Delete Menu Item 1. Manager or staff selects an existing menu item 2. Manager or staff selects the Delete Menu option 3. System requests confirmation 4. Manager or staff confirms deletion 5. System removes the menu item from the database 6. System displays a successful message 7. Use case ends

<table>
<tr><td>Exceptions</td><td colspan="3">E.1 Invalid Menu Input

1. System detects invalid or incomplete menu details
2. System displays an error message
3. Manager or staff re-enters menu details</td></tr>
<tr><td></td><td colspan="3">E.2 System Failure

1. System fails to save or delete menu data
2. System displays an error message
3. Use case is terminated</td></tr>
<tr><td>Related requirement</td><td>ID</td><td>Requirement</td><td>Priority</td></tr>
<tr><td></td><td>FR01-01</td><td>The system shall allow manager and staff to manage menu items</td><td>Basic</td></tr>
<tr><td></td><td>FR01-02</td><td>The system shall allow adding new menu items</td><td>Basic</td></tr>
<tr><td></td><td>FR01-03</td><td>The system shall validate menu input before saving</td><td>Performance</td></tr>
<tr><td></td><td>FR01-04</td><td>The system shall validate menu input before saving</td><td>Performance</td></tr>
<tr><td>Constraint Requirements</td><td>CR01-01</td><td>The system shall only authorized users shall manage menu items</td><td>Basic</td></tr>
<tr><td></td><td>CR01-02</td><td>The system shall require confirmation before deleting menu items</td><td>Performance</td></tr>
<tr><td>Quality Requirements</td><td>QR01-01</td><td>The system shall respond within 3 seconds during menu management</td><td>Performance</td></tr>
<tr><td></td><td>QR01-02</td><td>The system shall ensure menu data consistency</td><td>Basic</td></tr>
</table>

2.1.1 Activity Diagram of Manage Menu

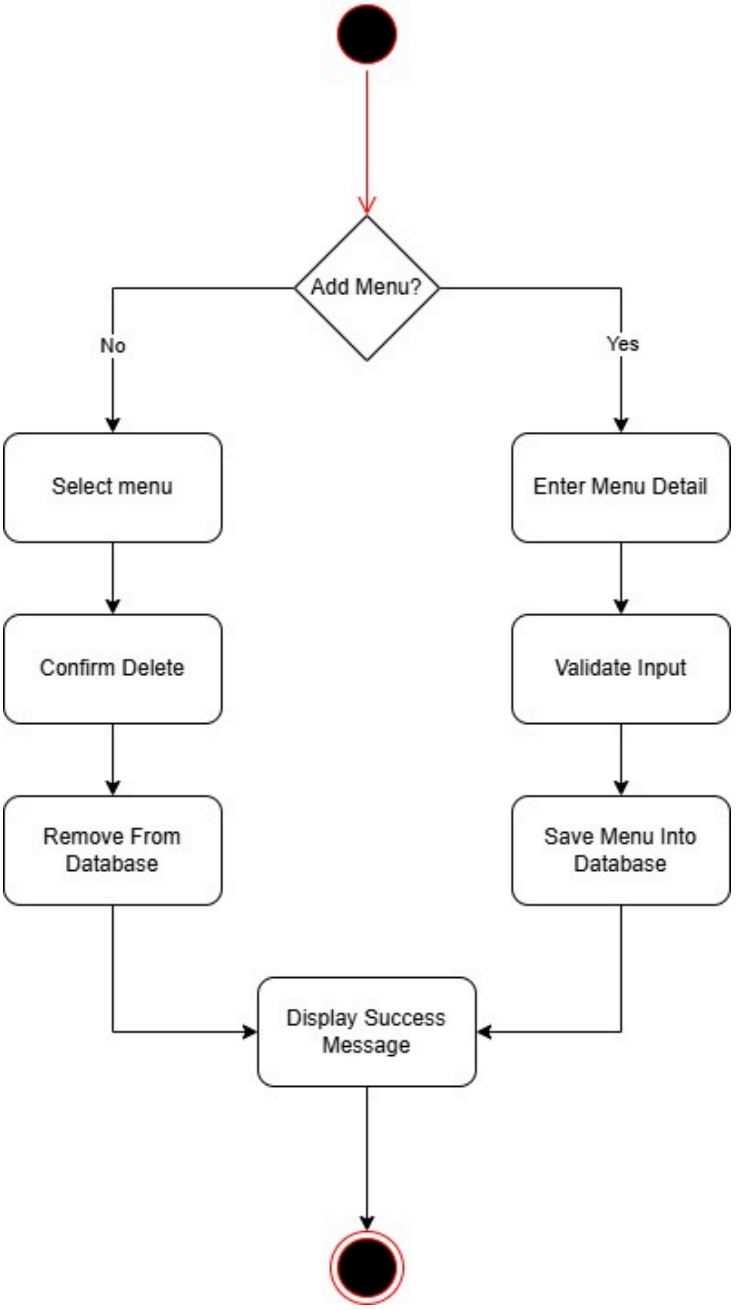


Figure 2.1.1: Manage Menu activity diagram

2.1.2 Sequence Diagram of Manage menu

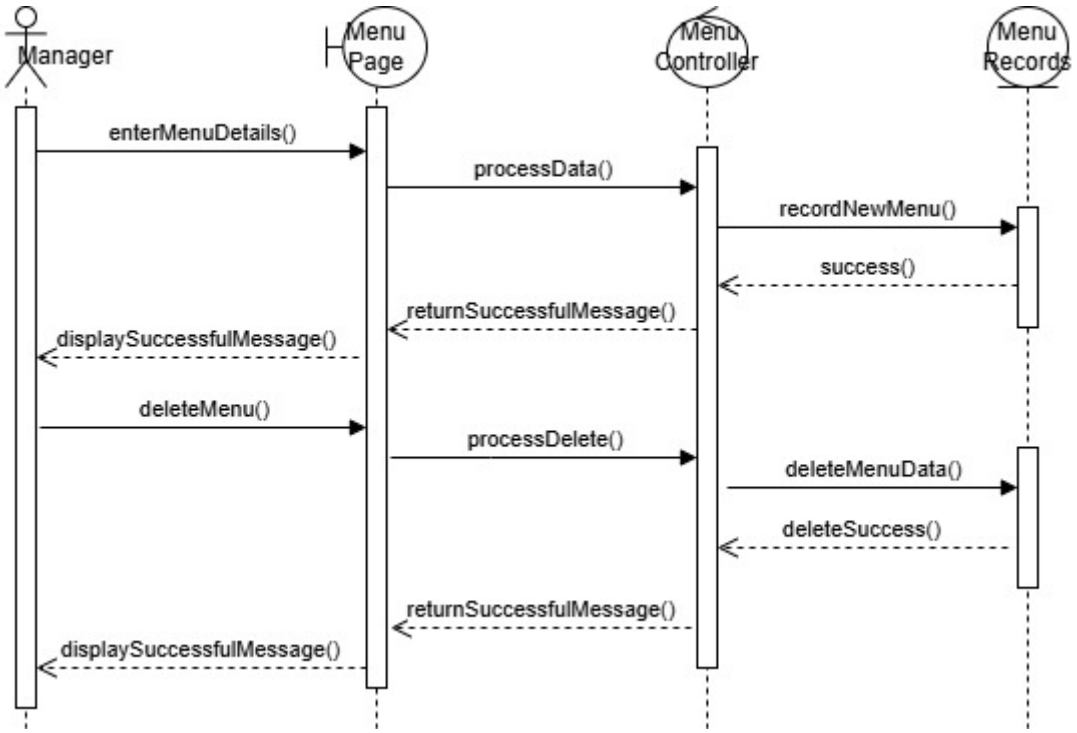


Figure 2.1.2: Manage Menu Sequence Diagram

2.2 Manage order use case specification

History Log	1.0.0	Initial creation of Manage Order use case	
	1.1.0	Added alternative flow for item availability	
	2.0.0	Updated actors and flows to align with sequence diagram	
Version	2.0.0		
Use Case ID	UC-2		
Use Case Name	Manage order		
Created By	Muhammad Faiz Ikhwan Bin Kamarudin	**Updated By**	Muhammad Amirul Hafiz bin Jasmani
Date Created	1/15/2026	**Last Revision Date**	1/17/2026
Actors	Customer, Staff, Kitchen		
Description	This use case outlines the process of an order being placed by the customer using the Azura House Automation System and the digital menu via the use of the QR code. Availability of the items, order ID, and coordination of the order details are done within the system before directing the customer to the payment process.		

Preconditions	1. 1.Customer has access to a mobile device with internet connection 2. 2.Customer has scanned the QR code successfully 3. 3.The system is online and operational
Post conditions	1. The order is successfully recorded in the system 2. A unique Order ID is generated 3. Order information is sent to staff and kitchen 4. Customer is redirected to the payment process
Normal Flow	**1.0 Manage Order** 1. Customer scans the QR code 2. System displays the digital menu 3. Customer selects food items and quantities 4. Customer confirms the order 5. System checks item availability 6. System generates a unique Order ID 7. System sends order information to staff and kitchen 8. System redirects the customer to the payment process 9. Use case ends
Alternative flow	A.1 Item Not Available 1. System detects that one or more selected items are unavailable 2. System displays an unavailable item message 3. Use case ends
	A.2 Item Not Available 1. System detects selected item is sold out of stock 2. System displays unavailable item message 3. Customer updates order selection
Exceptions	E.1 System Failure 1. System fails to process or save the order 2. System displays an error message

	3. Use case is terminated		
Related requirement	**ID**	**Requirement**	**Priority**
	FR02-01	The system shall be able to show a digital menu to the customer	Basic
	FR02-02	The system shall allow customer to select food items and quantities	Basic
	FR02-03	The system shall validate menu item availability	Performance
	FR02-04	The system shall generate a unique Order ID	Excitement
	FR02-05	The system shall redirect customer to payment after order confirmation	Basic
	FR02-06	The system shall notify staff and kitchen of a new order	Basic
Constraint Requirements	CR02-01	The customer shall not proceed without confirming the order	Basic
	CR02-02	The system shall not accept orders for unavailable items	Performance
Quality Requirement	QR02-01	The system shall respond within 3 seconds during order placement	Performance
	QR02-02	The system shall ensure data consistency for order records	Basic

2.1.1 Activity Diagram of Manage Order

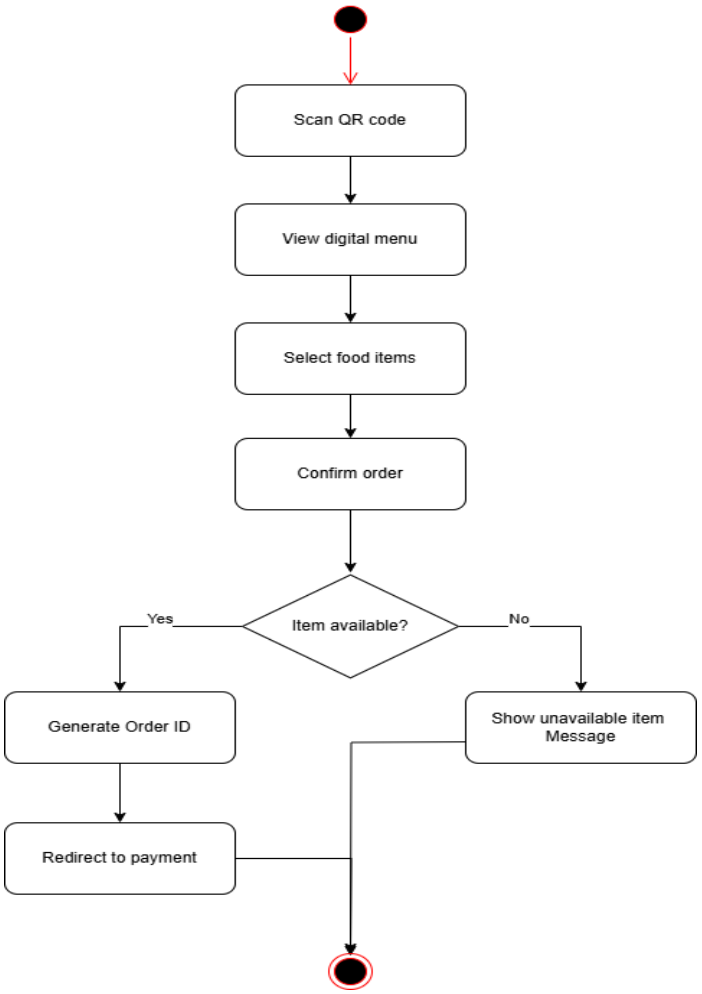


Figure 2.1 : Manage Order activity diagram

2.1.2 Sequence Diagram of Manage order

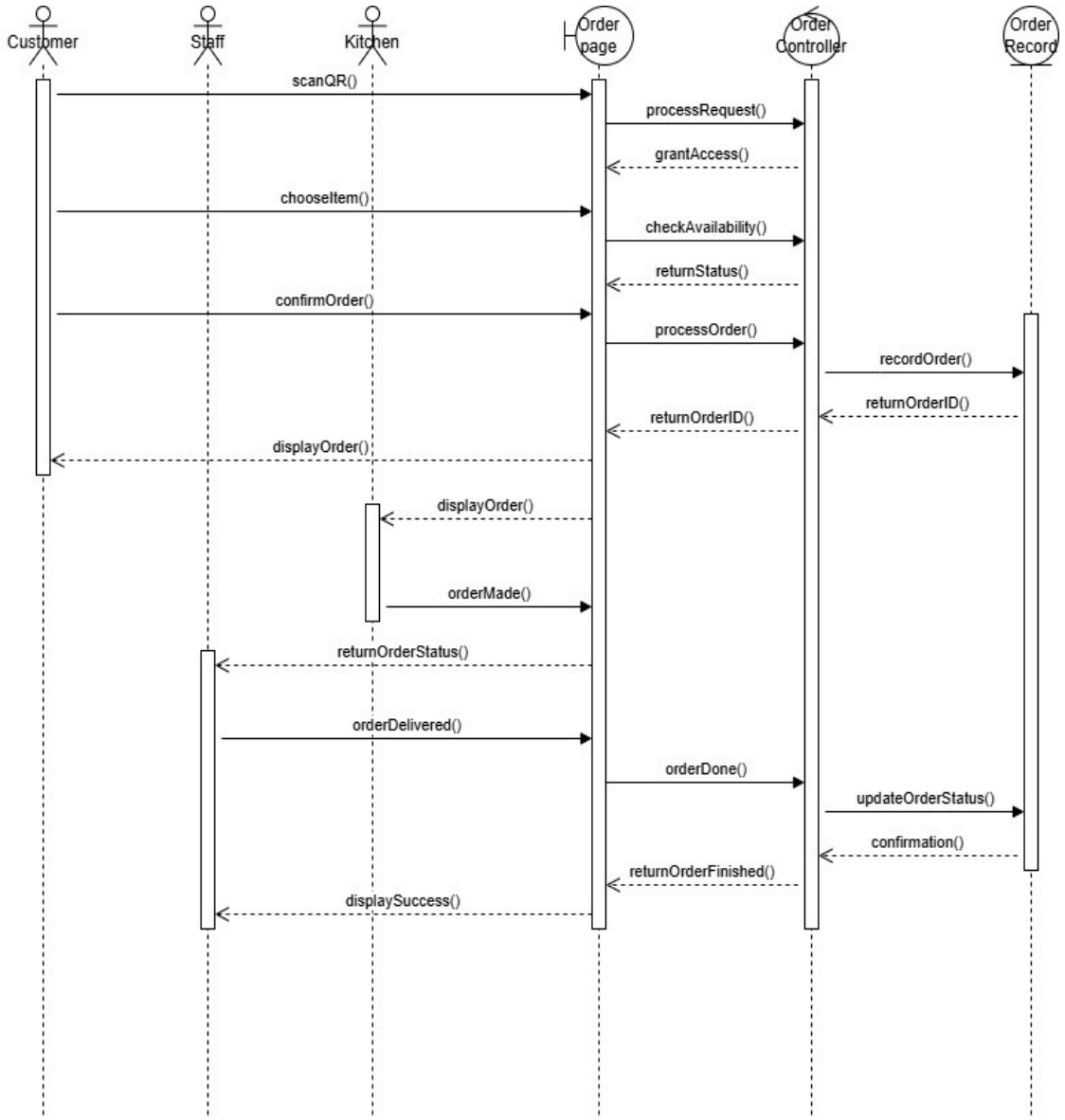


Figure 2.2: Manage Order Sequence Diagram

2.3 Manage Payment use case specification

History Log	1.0.0	Initial creation of Manage Payment use case	
	1.1.0	Added alternative flows for cash and online payment	
	2.0.0	Updated flows to align with activity and sequence diagrams	
Version	2.0.0		
Use Case ID	UC-3		
Use Case Name	Manage Payment		
Created By	Muhammad Amirul Hafiz Bin Jasmani	**Updated By**	Muhammad Faiz Ikhwan Bin Kamarudin
Date Created	1/14/2026	**Last Revision Date**	1/17/2026
Actors	Customer, Staff, Manager		
Description	Payment in the Azura House Automation System use case focuses on how payment is facilitated in the system. The scenario goes like this; the customer makes payment either in cash or online, and the system records this payment and generates sales records that are then reviewed by the manager in the form of the payment report.		
Preconditions	1. Customer has successfully placed an order 2. The Azura House Automation System is online and operational 3. Staff is available to handle cash payment if selected		
Post conditions	1. Payment transaction is successfully recorded 2. Sales record is updated 3. Payment receipt is generated and provided 4. Payment report is available for manager review		
Normal Flow	1.0 Manage Payment (Online Banking) 1. Customer selects online banking as the payment method 2. System redirects customer to the online banking payment page		

<table>
<tr><td></td><td colspan="3">3. Customer completes the payment transaction
4. System verifies the payment
5. System records the transaction
6. System updates the sales record
7. System generates a payment report
8. Manager views the payment report
9. Use case ends</td></tr>
<tr><td>Alternative flow</td><td colspan="3">A.1 Cash Payment

1. Customer selects cash as the payment method
2. Staff receives cash payment from customer
3. Staff enters payment details into the system
4. System records the transaction
5. System updates the sales record
6. System generates a payment report
7. System generates and provides receipt to customer
8. Manager views the payment report
9. Use case ends</td></tr>
<tr><td rowspan="2">Exceptions</td><td colspan="3">E.1 Payment Verification Failure

1. System fails to verify online payment
2. System displays an error message
3. Customer retries payment or selects another payment method</td></tr>
<tr><td colspan="3">E.2 System Failure

1. System fails to record payment transaction
2. System displays an error message
3. Use case is terminated</td></tr>
<tr><td>Related requirement</td><td>ID</td><td>Requirement</td><td>Priority</td></tr>
</table>

	FR03-01	The system shall allow customers to choose a payment method	Basic
	FR03-02	The system shall support cash payment processing	Basic
	FR03-03	The system shall support online banking payment	Basic
	FR03-04	The system shall record payment transactions	Performance
	FR03-05	The system shall generate payment receipts	Basic
	FR03-06	The system shall generate payment reports for manager	Basic
Constraint Requirements	CR03-01	Only staff shall handle cash payment processing	Basic
	CR03-02	Payment shall be verified before recording	Performance
Quality Requirements	QR03-01	The system shall respond within 5 seconds during payment method choosing	Performance
	QR03-02	The system shall ensure data recorded after successful payment	Basic

2.3.1 Activity Diagram of Manage Payment

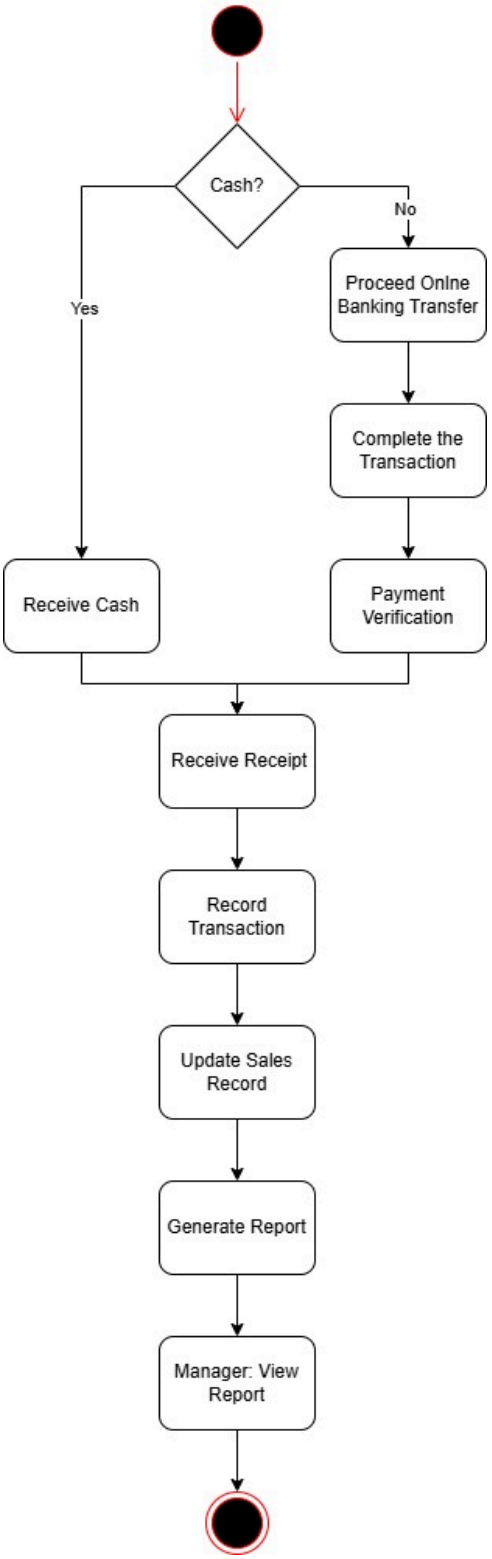


Figure 2.3.1 : Manage Payment Activity Diagram

2.3.2 Sequence Diagram of Payment Gateway

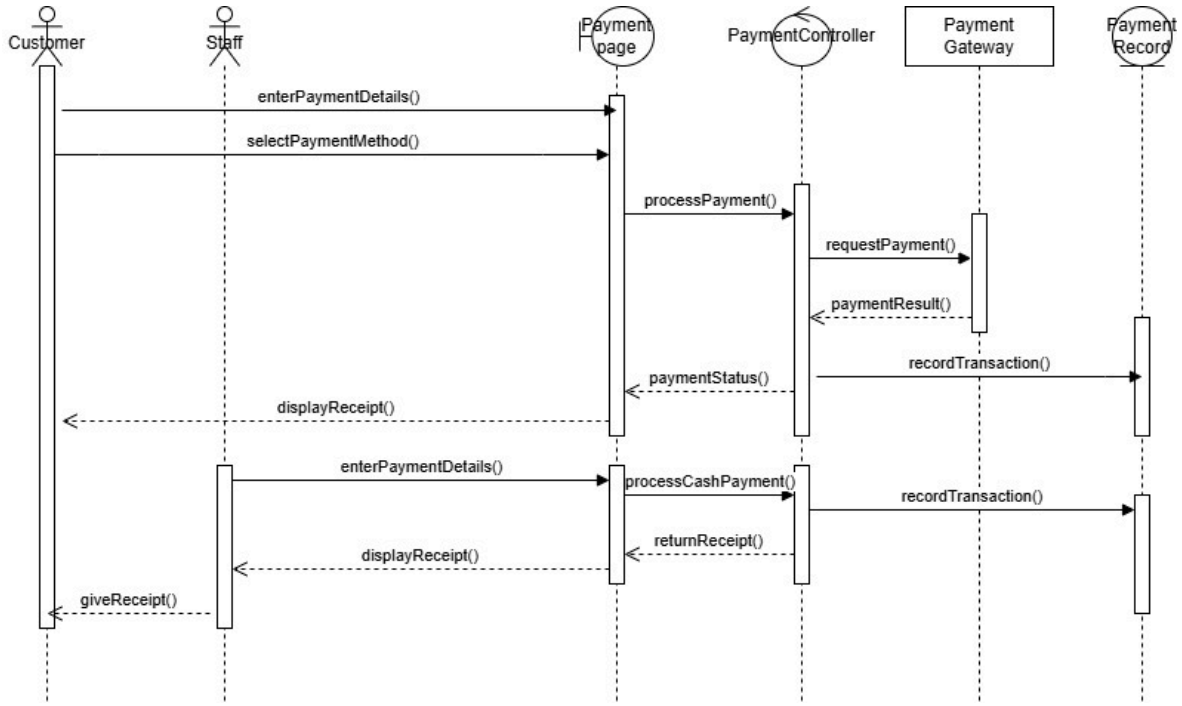


Figure 2.3.2 : Payment Gateway Sequence Diagram

2.4 Manage Inventory use case specification

History Log	1.0.0	Initial creation of Manage Inventory use case	
	1.1.0	Added alternative flows and exceptions	
	2.0.0	Updated actors, flows, and requirements	
Version	2.0.0		
Use Case ID	UC-4		
Use Case Name	Inventory Manage		
Created By	Muhammad Amirul Hafiz Bin Jasmani	**Updated By**	Afiq Aizad Bin Mazumi
Date Created	11/19/2025	**Last Revision Date**	17/01/2026
Actors	Customer		
Description	This use case describes how inventory items are monitored, updated, and managed in the Azura House Automation System. The system automatically detects low stock levels, notifies the manager,		

	and allows authorized staff to update inventory records when goods are received
Preconditions	1. Manager or staff has successfully logged into the system 2. The system is online and connected to the inventory database 3. User has authorization to manage inventory
Post conditions	1. Inventory records are updated successfully 2. Stock levels reflect real-time changes 3. Low-stock alerts are logged or resolved
Normal Flow	1. System continuously monitors inventory levels 2. System detects that an item has reached the minimum stock threshold 3. System sends a low-stock alert notification to the Manager 4. Manager reviews inventory details 5. Manager approves restocking action 6. Staff receives new goods from the supplier 7. Staff updates inventory status as “Received” System updates inventory records automatically Use case ends
Alternative flow	**A.1 Stock is sufficient** 1. System detects stock level is above minimum threshold 2. No alert is generated 3. Use case ends
	A.2 Sensor fault/system failure 1. Staff manually updates inventory quantities 2. System validates input 3. Inventory records are updated
Exceptions	**E.1 System Failure** **E.1 Invalid Inventory Input** 1. System detects invalid or incomplete inventory data 2. System displays an error message

	3. Staff re-enters correct information **E.2 System Failure** 1. System fails to update inventory records 2. System displays an error message 3. Use case is terminated		
Related requirement	**ID**	**Requirement**	**Priority**
	FR01-01	The system shall automatically detect when stock levels fall below the minimum threshold.	Basic
	FR01-02	The system shall send a real-time notification/alert to the Manager when an item is low.	Basic
	FR01-03	The system shall generate procurement documents based on the low stock data.	Performance
	FR01-05	The system shall allow the Manager to review and approve the generated documents before sending.	Basic
	CR01-01	The system shall electronically transmit the order to the Supplier once approved.	Basic
	CR01-02	The system shall update inventory and sales records immediately after Staff confirms receipt of goods.	Performance
	QR01-01	An order shall not be sent to the Supplier without the Manager's prior check/approval.	Basic
	QR01-02	The system shall only allow Staff to mark goods as "Received".	Basic

2.3.1 Activity Diagram of Manage Inventory

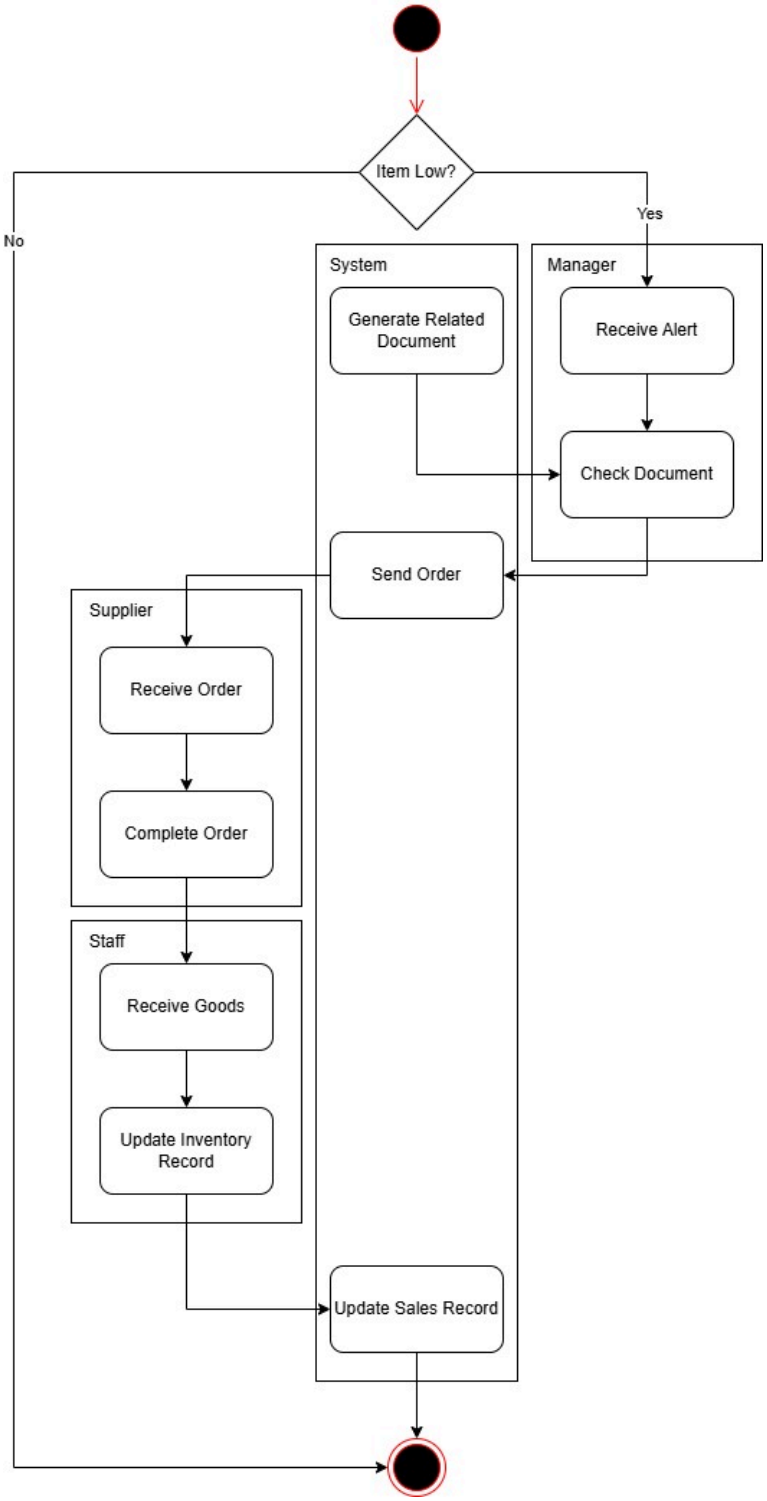


Figure 2.5 : Manage Inventory Activity Diagram

2.3.2 Sequence Diagram of Manage Inventory

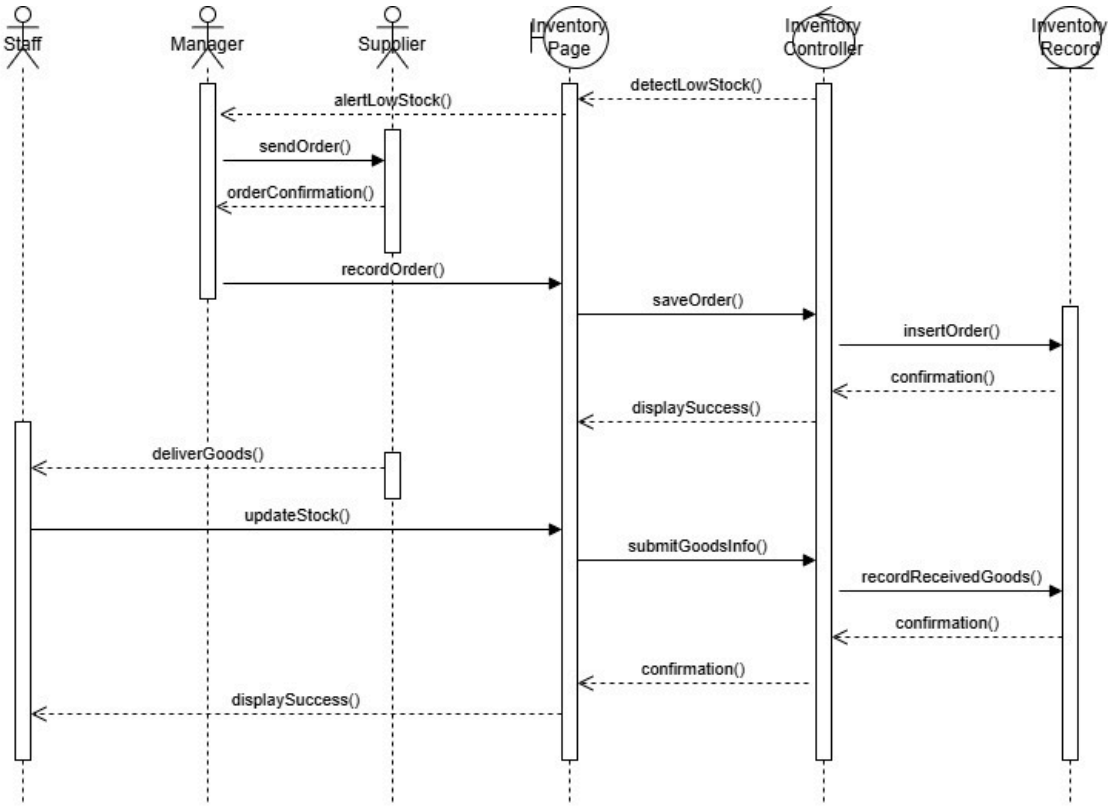


Figure 2.5 : Manage Inventory Sequence Diagram

2.5 Generate report use case specification

History Log	1.0.0	2. Initial creation of Inventory Manage use case	
	1.1.0	2. Added alternative flows and exceptions	
	2.0.0	2. Added functional, constraint, and quality requirements	
Version	2.0.0		
Use Case ID	UC-5		
Use Case Name	Generate Report		
Created By	Muhammad Amirul Hafiz Bin Jasmani	**Updated By**	Afiq Aizad Bin Mazumi
Date Created	11/19/2025	**Last Revision Date**	17/1/2026
Actors	Staff, Manager, Supplier		

Description	This use case describes how the manager generates system reports such as sales reports, payment reports, and inventory reports to support monitoring, analysis, and decision-making.
Preconditions	1. Manager has successfully logged into the system 2. System contains recorded order, payment, and inventory data 3. System is online and operational
Post conditions	1. Selected report is generated successfully 2. Report is displayed or downloadable 3. Data accuracy is maintained
Normal Flow	1. Manager selects the Generate Report option 2. System displays available report types 3. Manager selects report type (sales, payment, inventory) 4. Manager selects date range or filters 5. System retrieves relevant data 6. System generates the report 7. System displays the report to the Manager 8. Use case ends
Alternative flow	**A.1 Export Report** 1. Manager selects export option (PDF/CSV) 2. System generates downloadable file 3. File is saved or downloaded
Exceptions	**E.1 No Data Available** 1. System detects no data for selected criteria 2. System displays “No data available” message **E.2 System Failure** 1. System fails to generate report 2. System displays an error message

	3. Use case is terminated		
Related requirement	**ID**	**Requirement**	**Priority**
	FR04-01	The system shall generate sales reports	Basic
	FR04-02	The system shall generate inventory reports	Basic
	FR04-03	The system shall generate payment reports	Basic
	CR04-01	Only managers can generate reports.	Basic
	QR04-01	Reports shall be generated within 5 seconds	Performance
	QR04-02	Reports shall reflect accurate system data	Basic

2.5.1 Activity Diagram of Generate Report

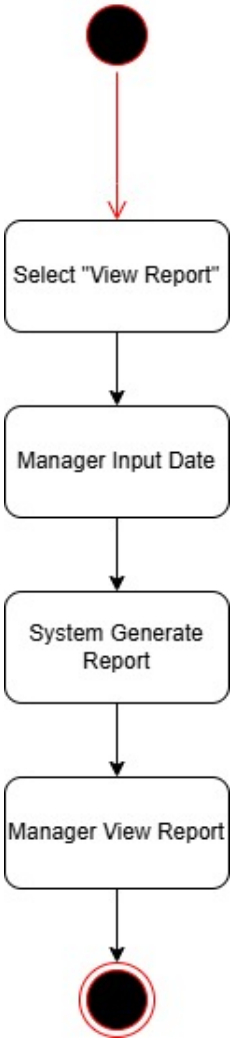


Figure 2.5 : Manage Inventory Sequence Diagram

2.3.2 Sequence Diagram of Manage Inventory

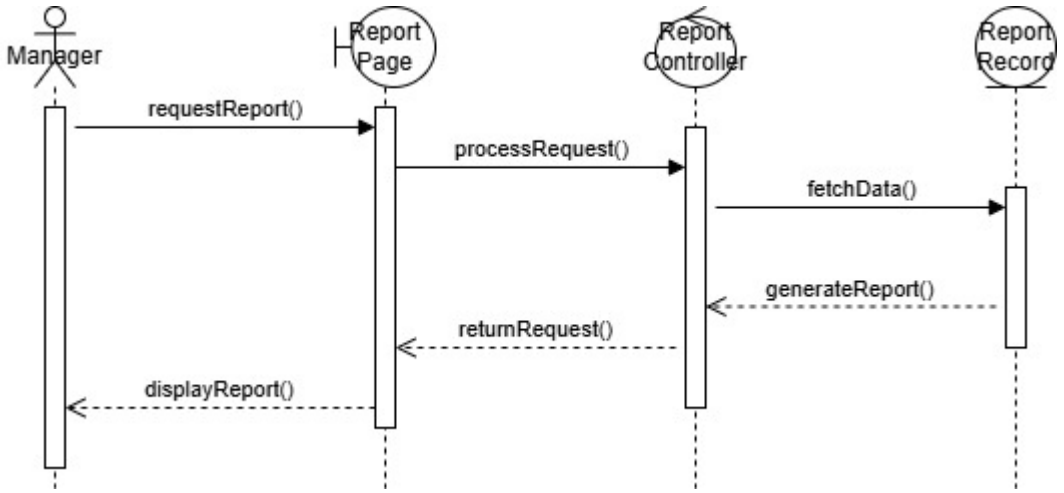


Figure 2.5 : Manage Inventory Sequence Diagram

2.6 Domain Class Diagram

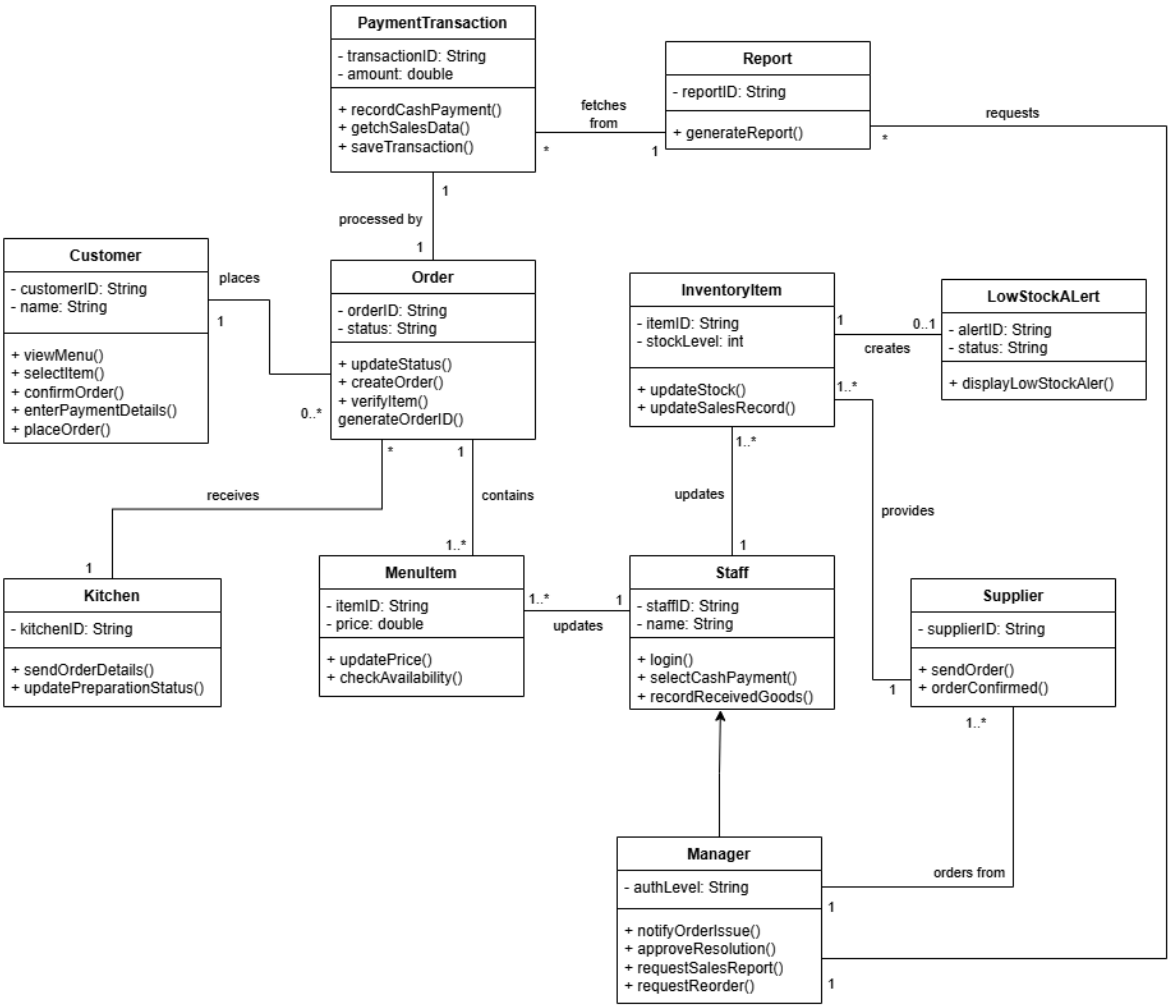


Figure 2.7 : Domain Class Diagram of Azura House

Business Rule

Ordering System Rules

1. Payment must be completed before the order is sent to the staff.
2. Menu items and prices must be controlled by an authorized staff.
3. Each order must have a unique ID.
4. The kitchen shall only receive an order that has been paid.
5. The system must perform an immediate check to ensure requested items are available before accepting an order.

Payment Module Rules

1. Customers must place an order before making any payment.
2. Customers must choose one valid payment method (e.g., QR, Cash, Online Transfer).
3. A transaction is recorded only after payment is successful.
4. Customers must receive a receipt after payment.
5. Staff can help customers but cannot bypass system verification.
6. All payment and transaction data must be securely stored.

Inventory Management Rules

1. Only authorized staff can perform receiving and document-checking tasks.
2. Low-stock items trigger an automatic supply check and may initiate an order request.
3. The system must generate related documents whenever inventory changes.
4. Stock levels must be updated automatically once goods are received or sales occur.
5. Ingredients must be deducted from the total stock every time kitchen staff use them (synced with customer order).

2.7 Requirement Traceability

Table 1: Requirements traceability matrix

Allocated	Descriptions
REQ_100	**Azura House Ordering System**
FR01-01	The system shall be able to show a digital menu to the customer
FR01-02	The system shall allow customers to select food items and quantities.
FR01-03	The system shall validate menu item availability
FR01-04	The system shall generate a unique Order ID.
FR01-05	The system shall redirect the customer to payment after order confirmation.
CR01-01	The customer shall not proceed without confirming the order.
CR01-02	The system shall not accept orders for unavailable items.
QR01-01	The system shall respond within 3 seconds during order placement.
QR01-02	The system shall ensure data consistency for order records.
REQ_200	**Azura House Payment Module**
FR02-01	The system shall allow customers to choose valid payment methods (e.g., QR, Cash, Online Transfer).
FR02-02	The system shall generate and issue a digital receipt after successful payment.
FR02-03	The system shall allow managers to generate daily sales reports for all payment types
CR02-01	Customers must place an order before making any payment.
CR02-02	A transaction shall only be recorded after a payment is confirmed as successful.
CR02-03	Staff shall be prohibited from bypassing system verification steps.
QR02-01	All payment and transaction data must be securely stored.
REQ_300	**Azura House Inventory Management Module**
FR03-01	The system shall update stock levels automatically once goods are received or sales occur
FR03-02	The system shall trigger automatic supply checks and low-stock alerts.
FR03-03	The system shall generate related documents (e.g., reports) whenever inventory changes
FR03-04	The system shall deduct ingredients from stock based on customer orders.
CR03-01	Receiving and document-checking tasks shall be restricted to authorized staff only.

CR03-02	Orders shall only be sent to the kitchen after required payment documents are validated
QR03-01	The system shall maintain real-time synchronization between kitchen usage and stock levels

3. Prototype Implementation

3.1 Prototype Overview

Azura House Restaurant Automation System prototype has been created to show how the requirements identified and described in Section 2 of this Software Requirements Specification (SRS) can be implemented. The prototype has come into existence as a functional prototype that indicates the completion of the implementation of system functionalities. Basically, the main purpose of creating the prototype is to ensure that the functional requirements and quality requirements are tested through a system that has come from models designed.

The prototype involves core restaurant processes such as taking customer orders through the digital menu system, payment through the payment gateway system, and management of the inventory system. It ensures system flow requirements are adhered to, such as completing payment transactions so that preparation can be done. It demonstrates how the system requirements, use cases, and system models have been implemented in a system.

Staff dashboard :

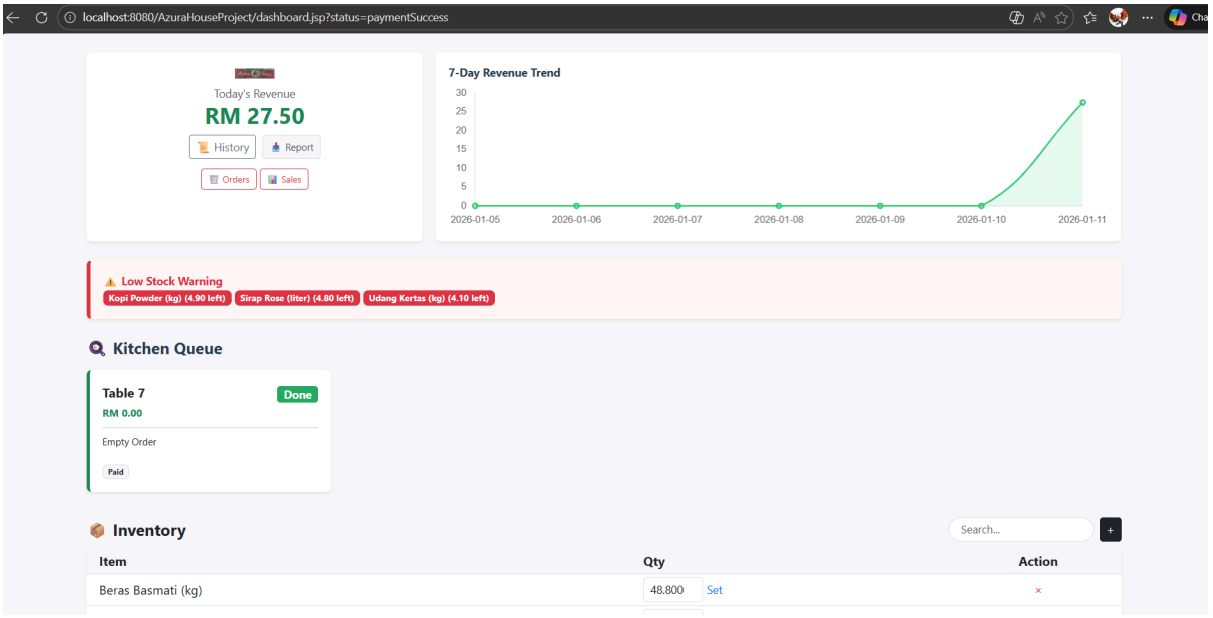


Figure 3.1.1 Staff Dashboard

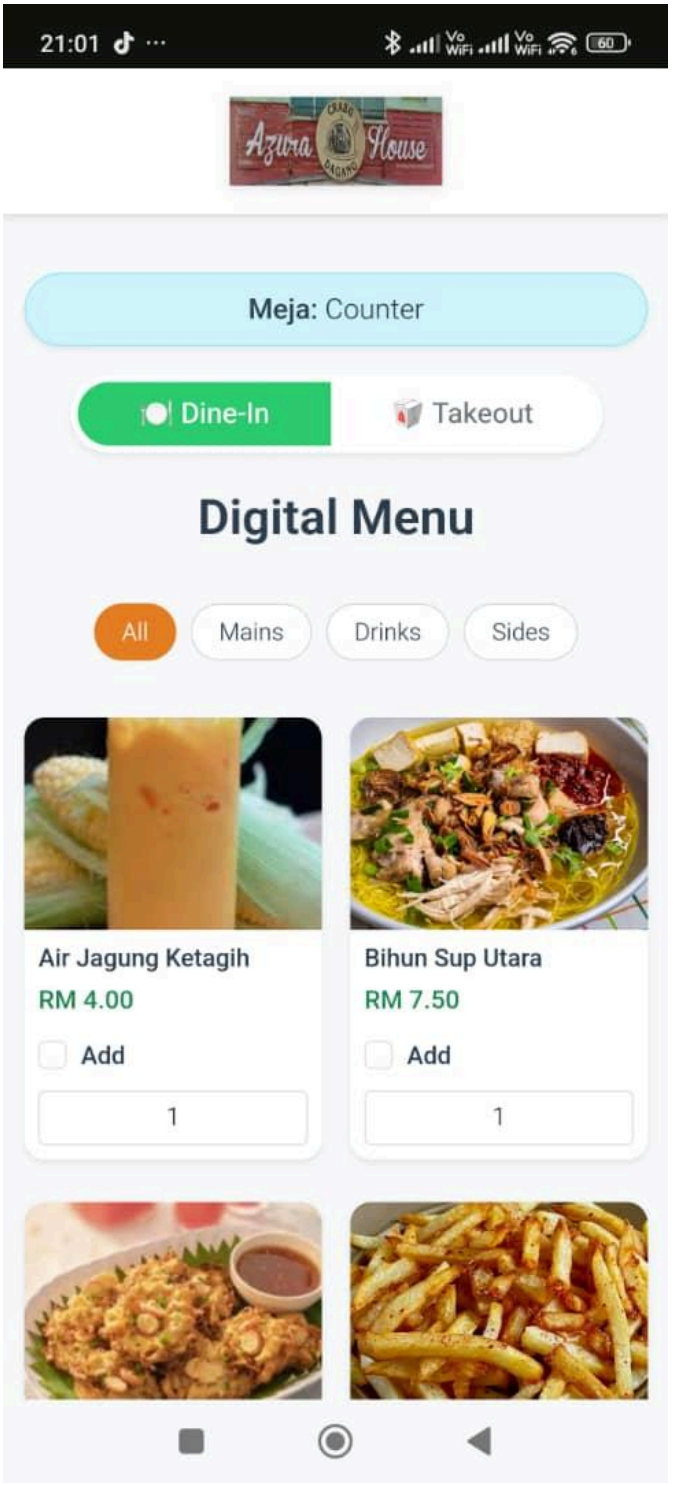


Figure 3.1.2 Order Page

3.2 Technology Used

The prototype was implemented using the following technologies:

- Java – Core programming language used to implement system logic and business rules
- Java Server Pages (JSP) – Used to create dynamic web pages and handle user interaction
- JSON – Lightweight data interchange format used to store and manage system data such as orders, payments, and inventory
- Apache Tomcat – Web application server used to deploy and run the Java web application
- HTML, Javascript & CSS – Used for user interface design and basic styling

These technologies were selected due to their reliability, scalability, and suitability for developing a web-based restaurant automation system.

3.3 Database Design

The data design on how the Azura House Restaurant Automation System would work came directly from the domain class diagram depicted in Section 2. Each domain class identified was translated into a structured data file, which stored system data that persisted. Instead, a file system implementation using JSON files would be used as a simplified alternative to a relational database system that still supported data integrity and requirement validation on the system.

Implemented Data Files:

• menu.json – Stores menu item information

• orders.json – Stores customer order records

• inventory.json – Stores inventory and stock data

For instance, the file orders.json contains data for orders that link to menu items, thus preventing the creation of an order based on data that does not exist within the menu entities. This is a logical association that enables appropriate process flow when the process involves, for instance, processing an order before making payments.

In this section, the design of the data implemented for the Azura House Restaurant Automation System shall be described. In designing the data structure for the project, the classes identified and the functional requirements identified in the Software Requirements Specification (SRS) had been used. Data entities were maintained in a structured format with designated attributes for integrity and consistency.

The design is capable of handling fundamental system functions such as ordering and payment. The relationship defined in a diagrammatic form is implemented in terms of validate methods in the java controller and DAO classes.

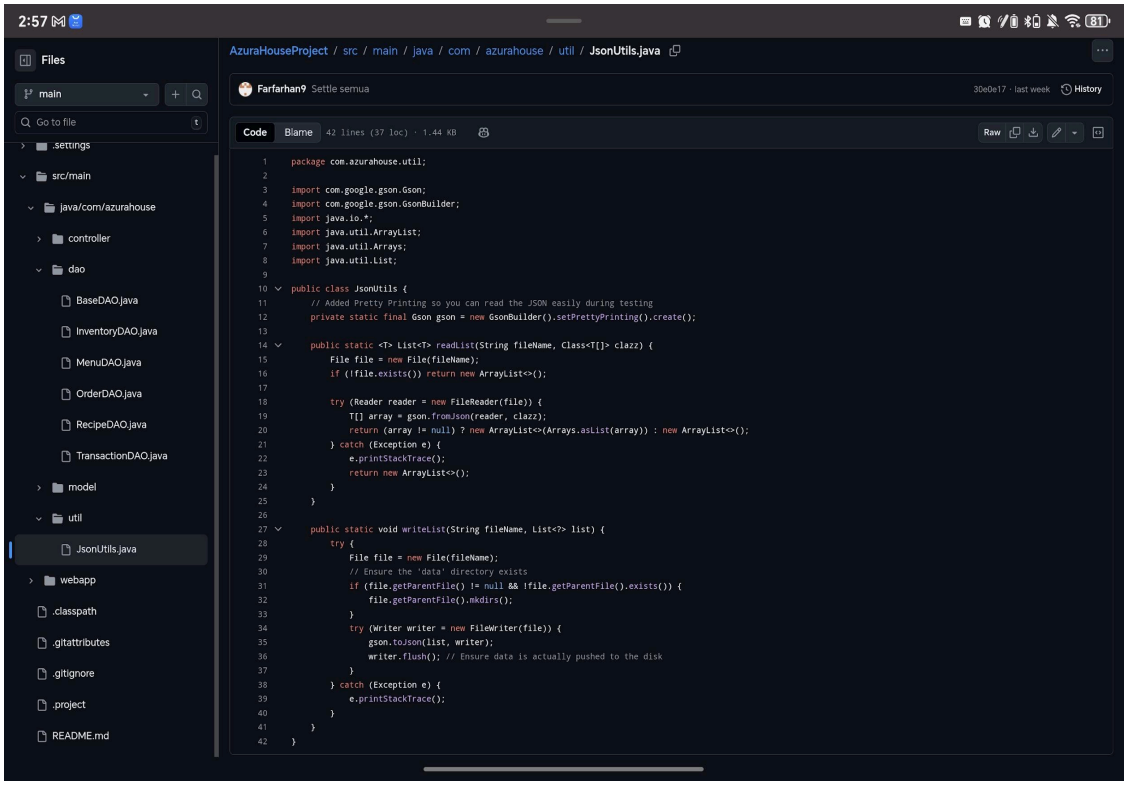


Figure 3.1: Data structure for Azura House Restaurant Automation System

3.4 Use Case to Prototype Mapping

The following table shows the traceability between use cases and their prototype implementation.

Use Case	Description	Prototype file
UC-1	Place Order	index.jsp OrderServlet.java
UC-2	Payment Gateway	payment.jsp, PaymentServlet.java
UC-3	Inventory Manage	dashboard.jsp InventoryServlet.java

This mapping demonstrates the transformation of conceptual use cases into executable system components.

3.5 Process Enforcement

The prototype will adhere to the proper order of tasks in the system to guarantee the system behaves logically. Some operations have been limited considering the end of prerequisite processes not to have illegal system usage and in line with the system requirements. For example, the payment process can only occur after the customer has placed an order, and changes can only occur when the system access of the authorized staff is gained.

This enforcement is carried out through application level validation in the Java Servlets and by the controller logic. Access to the payment page is denied if there is an invalid order within the current session, and the functionality for inventory management is secured to be accessible only through the staff dashboard page.

Below are screenshots that showcase functionality validation in terms of acts performed by a user and ensuring invalid access is not permitted in case the necessary steps are not followed. This is done to ensure that system functionality is as per the defined functionality and constraints.

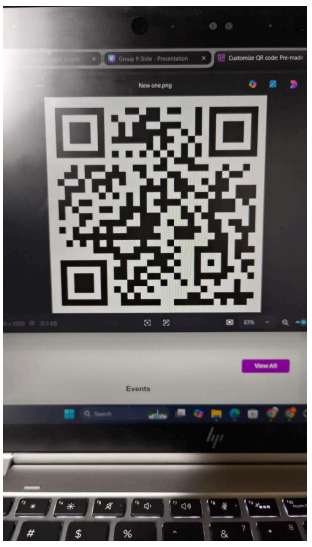


Figure 3.5.1 QR Code Menu

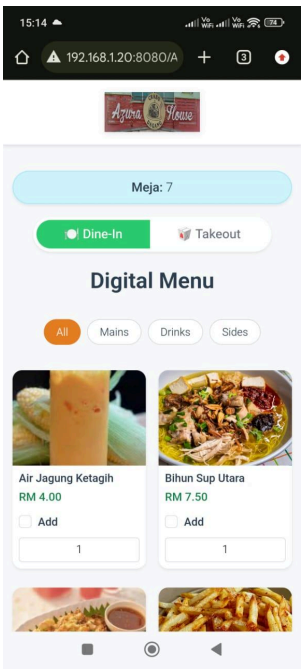


Figure 3.5. Menu Page (Customer)

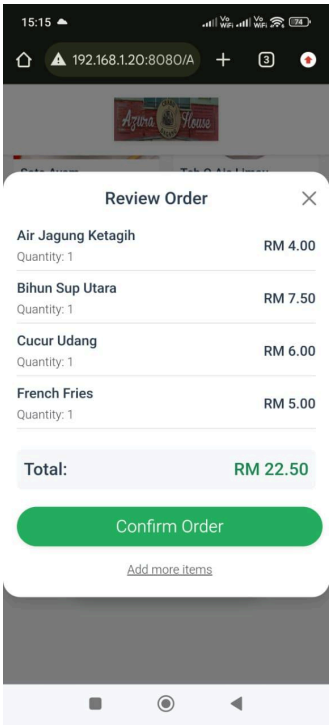


Figure 3.5.3 Confirm Order Overlay

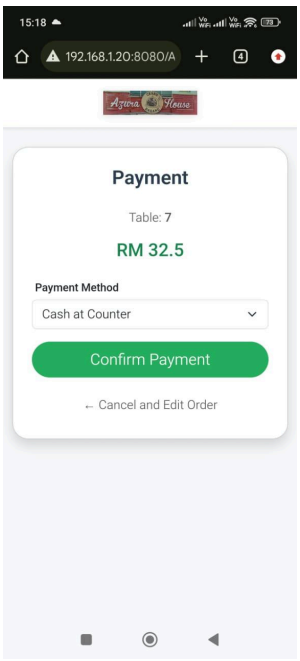


Figure 3.5.4 Payment Page

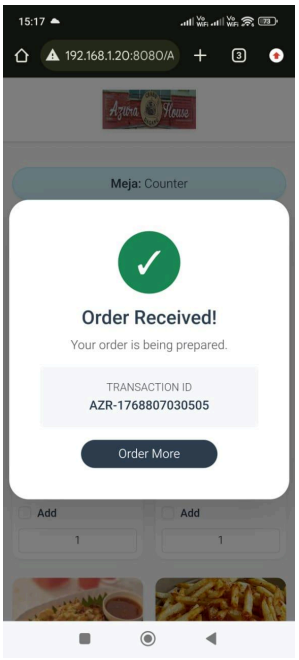


Figure 3.5.5 Payment Completed Overlay

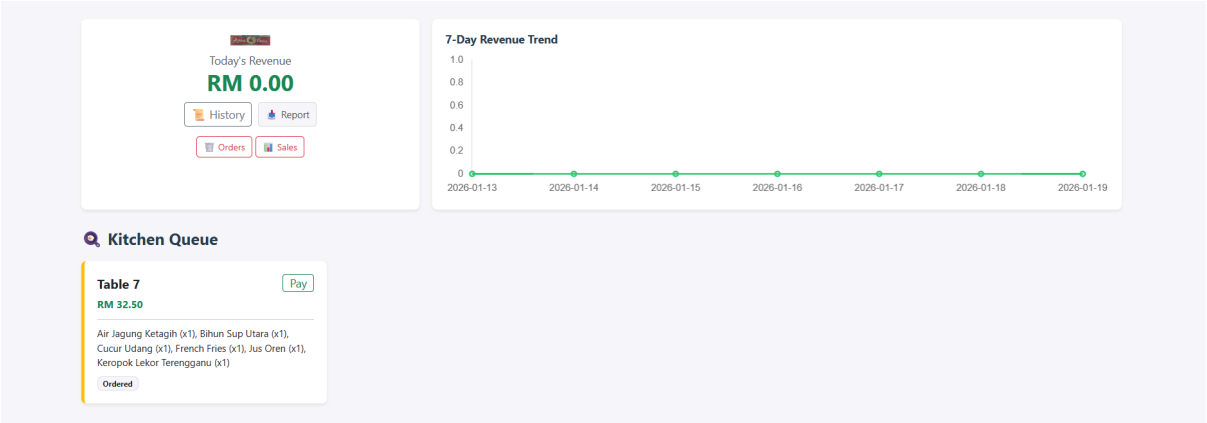


Figure 3.5.6 Staff Dashboard (Revenue and Kitchen Queue Section)

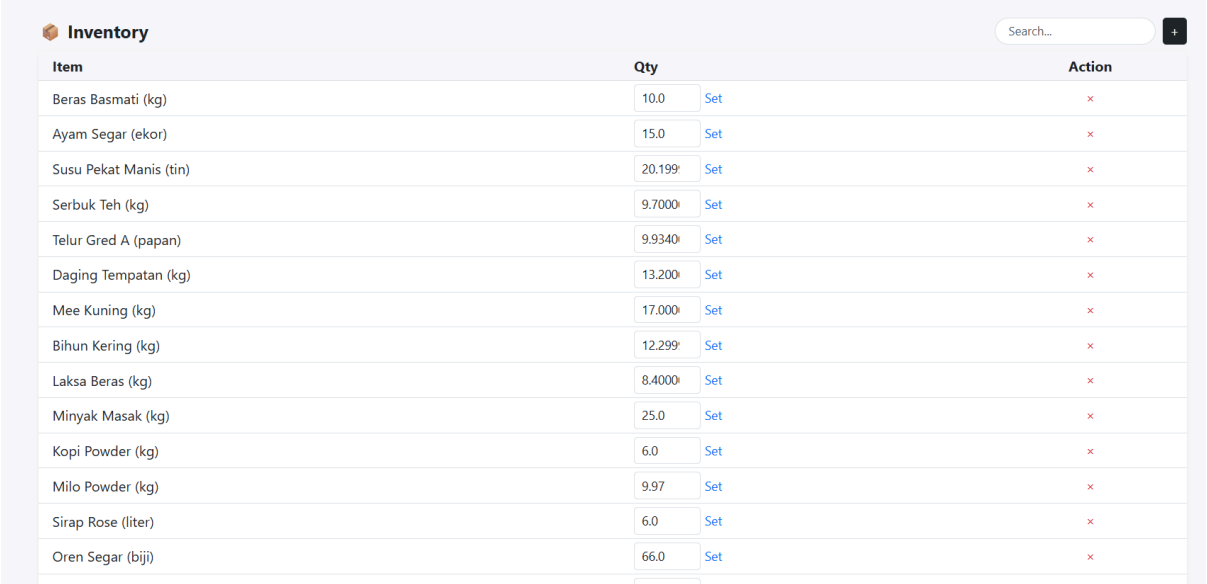


Figure 3.5.7 Staff Dashboard (Inventory Section)



Figure 3.5.8 Staff Dashboard (Recipe Linker Section)

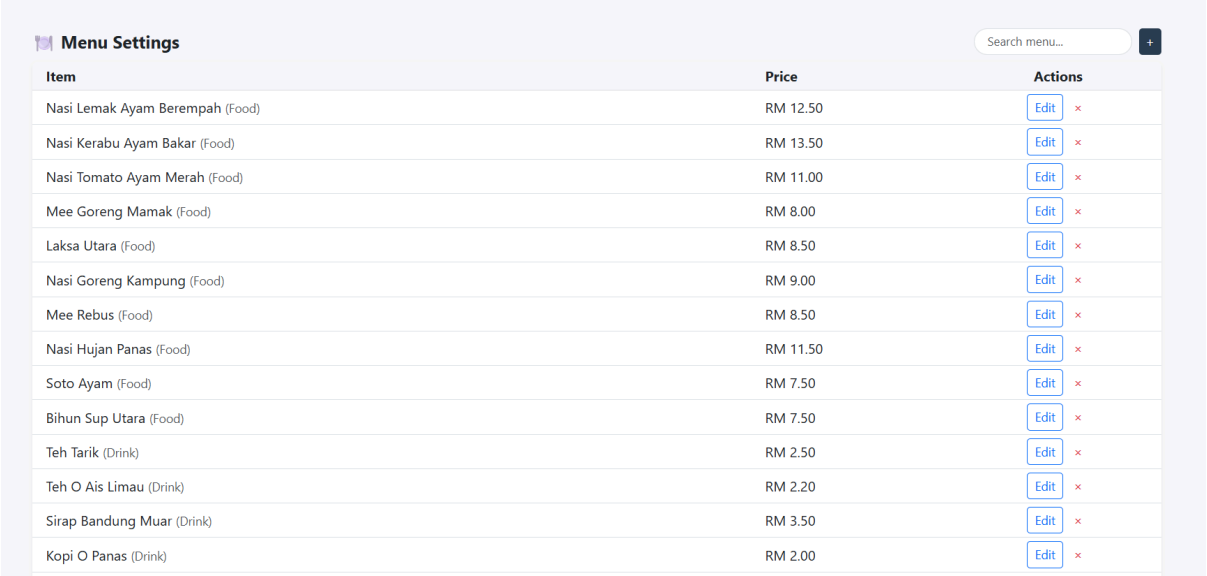


Figure 3.5.9 Staff Dashboard (Menu Settings Section)

4.0 Validation and Verification

This section explains how the Azura House Cafe Automation System requirements are validated and verified to ensure that the system meets user needs, follows defined constraints, and achieves the expected quality level. Validation ensures that the correct system is built, while verification ensures that the system is built correctly according to the specified requirements.

4.1 Requirement Validation Using Prototype

Requirement validation is performed using a functional prototype of the Azura House Cafe Automation System. The prototype demonstrates the main system modules, including the ordering module, payment module, and inventory management module.

Each major use case is validated as follows:

- Place Order Use Case: The prototype allows customers to scan a QR code, view the digital menu, select food items, confirm orders, and receive an Order ID. This confirms that functional requirements such as menu display, item selection, availability checking, and order confirmation are correctly implemented.
- Payment Gateway Use Case: The prototype shows customers choosing a payment method, completing payment, and receiving payment confirmation. This validates

that payment selection, payment validation, and Payment ID generation meet the defined requirements.

- Inventory Management Use Case: The prototype demonstrates automatic low-stock detection, alert notifications to managers, document generation, and inventory updates after goods are received. This validates that inventory-related requirements support real operational scenarios.

Feedback from simulated users (customer, staff, and manager roles) confirms that the system behavior matches the intended business processes defined in the SRS.

4.2 Constraint Enforcement

Constraint enforcement ensures that system rules and limitations are strictly followed during system operation. The prototype enforces constraints in the following ways:

- Customers are not allowed to proceed to payment without confirming their order.
- Orders for unavailable or out-of-stock items are rejected by the system.
- Payments cannot be made without an existing order.
- Only authorized staff and managers can access inventory management and approval functions.
- Orders are only sent to the kitchen after successful payment verification.

These constraints are enforced through system validations, role based access control, and confirmation checks within the prototype. This ensures compliance with constraint requirements defined in the ordering, payment, and inventory modules.

4.3 Quality Requirement Considerations

Quality requirements focus on system performance, reliability, usability, and data consistency. The following quality aspects are considered:

- Performance: The prototype is designed to respond within the specified time limits, such as displaying menus and processing payments within a few seconds.
- Reliability: Error messages are displayed when system failures occur, such as unsuccessful payments or inventory update issues.
- Usability: The user interface is simple and intuitive, allowing customers to place orders and make payments with minimal steps.
- Data Consistency: Order, payment, and inventory data are synchronized to ensure accurate records across all modules.
- Security: Payment and transaction data are protected through system validation and controlled access.

These quality considerations ensure that the system provides a smooth and trustworthy user experience.

4.4 Verification Summary

Verification activities confirm that the system implementation aligns with the specified requirements in the SRS. This is achieved through:

- Reviewing each functional requirement against prototype features.
- Checking that all constraints are enforced correctly during system operation.
- Ensuring quality requirements such as response time and data accuracy are met.
- Validating use case flows using activity and sequence diagrams as references.

Overall, the verification results show that the prototype satisfies the functional, constraint, and quality requirements defined for the Azura House Cafe Automation System. This confirms that the system is ready to proceed to further development or evaluation stages.

5.0 Conclusion

This Software Requirements Specification (SRS) has successfully defined the complete functional, constraint, and quality requirements for the Azura House Cafe Automation System. By using structured requirement modeling, detailed use case specifications, activity and sequence diagrams, and business rules, this document provides a clear description of how the system should operate to support restaurant ordering, payment, and inventory management processes.

The development of a functional prototype further strengthened the requirements by demonstrating how the proposed system can be implemented in practice. The prototype validated that customers would be able to place orders through a QR-based digital menu, complete payments through multiple payment methods, and that inventory levels are automatically updated and monitored. The mapping between use cases and prototype components confirmed that all key requirements have been translated into working system functions.

Validation and verification activities showed that the system meets its defined constraints, including preventing payments without orders, blocking unavailable items, and restricting inventory access to authorized staff. In addition, quality requirements such as performance, data consistency, usability, and security were considered and implemented within the prototype, ensuring the system is reliable and suitable for real-world restaurant operations.

Overall, the Azura House Cafe Automation System is well-defined, validated, and ready for further development or deployment. The SRS and prototype together provide a solid foundation for building a fully operational restaurant automation system that improves efficiency, reduces errors, and enhances customer experience.

6.0 References

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